MAI1202 Assignment 2

The work that you complete and submit must be your own, not the work of others, nor the creation of an AI tool. It is permissible to discuss the assignment with others in general terms, discuss general approaches to problems, help someone to understand a concept or give them a bit of advice if they have a problem, etc. However, you are not permitted to 'work together', nor to give/receive solutions (whether partial or complete) to/from others.

If you are unclear at all about what is permissible, please ask your professor.

Late submissions will not be accepted.

Dataset

- https://www.kaggle.com/datasets/thedevastator/weather-prediction/data (weather_prediction_dataset)

Task

Your task is to build sequence models for the given dataset. More specifically, you will need to experiment with several architectures, comparing computation and performance for each below scenario.

Train-Test Split: 70-30

Scenario 1: Single-step (14 → 1)

- **Feature: BASEL_temp_mean**
- Slide a 14-day window, predict the **next 1 day**

Scenario 2: Multi-step (14 → 3)

- **Feature: BASEL_temp_mean**
- Slide a 14-day window, predict the **next 3 days at once**

Scenario 3:

- **Features: BASEL_temp_mean and BASEL_humidity**
- Slide a 14-day window, predict the **next 3 days at once**

Scenario 4:

Use the following input features:

- `BASEL_temp_mean`
- `BASEL_humidity`
- `MAASTRICHT_temp_mean`
- `DUSSELDORF_temp_mean`
- `MUENCHEN_temp_mean`
- `DE_BILT_temp_mean`
- `MAASTRICHT_humidity`

- `DUSSELDORF_humidity`
- `MUENCHEN_humidity`
- `DE_BILT_humidity`

The prediction targets remain:

- `BASEL_temp_mean`
- `BASEL_humidity`

- **Slide a 14-day window, predict the next 3 days at once**

For each scenario, do these steps:

1. Prepare the dataset properly for machine learning, considering both the problem and the types of models you will use.
2. Build the following types of models, applying appropriate practices, and evaluate performance/training time for each:
    a. A simple RNN
    b. A more elaborate RNN (e.g., deeper)
    c. A LSTM network
    d. A GRU network
3. Compare the performance/training time for each of these models.
4. Reflect on the results you obtained: explain the relative performance of the models you tested on this problem and dataset.

Requirements

Your submission should consist of a single notebook file, containing both your report text and your code/output. (The two should be integrated appropriately; do not put all of the text at the beginning/end.)

Submit both your Notebook file, as well as a PDF file of your notebook.

Important note: You may be interviewed or quizzed about your submission, to ensure your understanding of the work submitted. Your ability to demonstrate understanding will be factored into your grade.